Virtualization
Operating Systems – EDA093/DIT401

Vincenzo Gulisano
vincenzo.gulisano@chalmers.se

Reading instructions

- Chapter 7, 7.1-7.10 (included)

Agenda

- Introduction
- Requirement for virtualization
- Type 1 and type 2 hypervisors
- Techniques for Efficient Virtualization
- Memory virtualization
- I/O virtualization
- Other benefits of virtualization
- Clouds
- Containers

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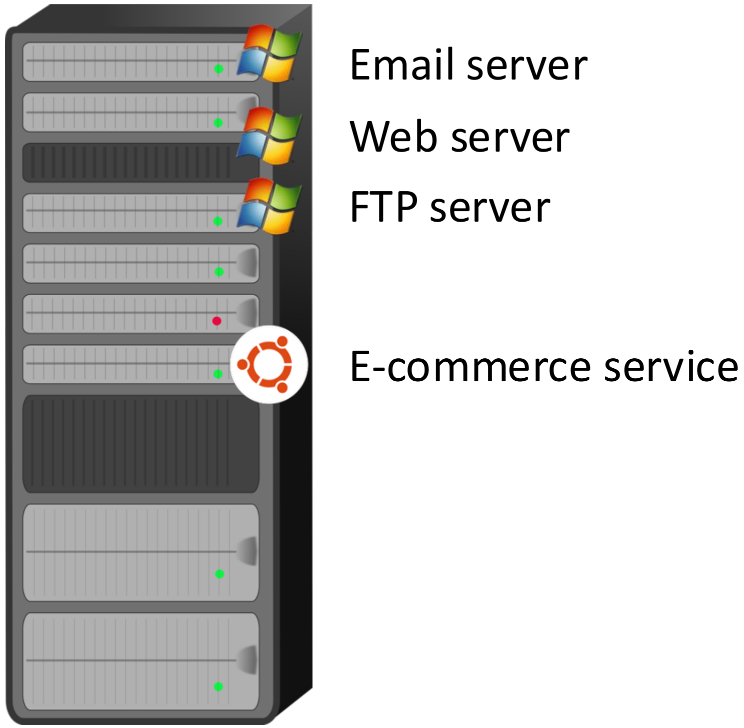
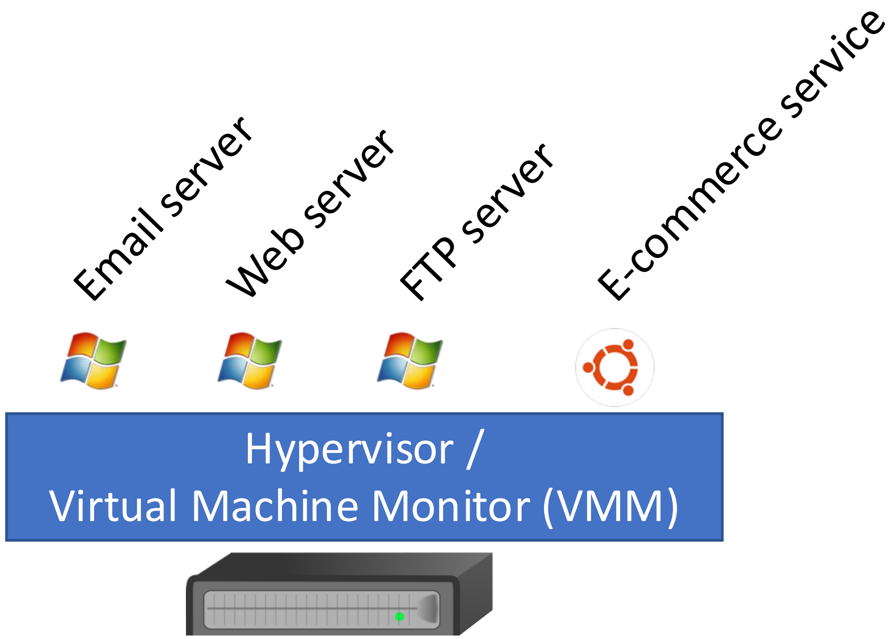
Without virtualization



With virtualization
(discussed since the 60s…)



	Without virtualization	With virtualization
PROS	• If one machine cannot handle the entire load, it works • Reliability (1 server crash does not imply all services are down) • Better for security (sandboxing) – if an intruder compromises web server, does not have access to mails • Run different OSes • …	• If the failures are at software/OS level, still reliable • Run different OSes • Hypervisor code (which need to be the one you can really trust) is smaller than OS and easier to maintain • Costs less • Cheaper to try out applications/services/… (good in software development, to test with different OSes) and provide **virtual appliances** • Legacy applications • Easier to migrate services/OSes
CONS	• Costs money and time	• Hardware failure can have catastrophic consequences

As you can imagine, the Cloud starts from this idea and takes it to that of using a server located somewhere else, which further reduces costs…

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Aside from “acting like a physical machine”, what are actually the requirements?

- Safety (hypervisor should have full control of the virtualized resources)
- Fidelity (program running on bare hardware = program running on a VM)
- Efficiency

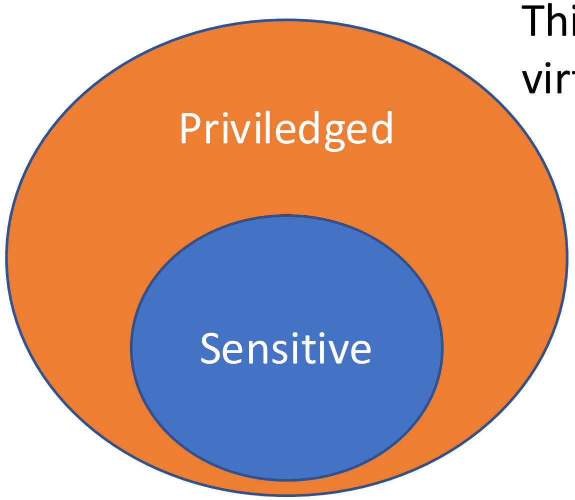
Why can’t we ”just” use an interpreter?

- Wrap each instruction
  - Execute if safe (e.g., INC instruction)
  - Simulate otherwise (e.g., disable interrupts)
- Theoretically it works, but
  - Safety ✓
  - Fidelity ✓
  - Efficiency ✗
- VMMs should execute most of the code directly…

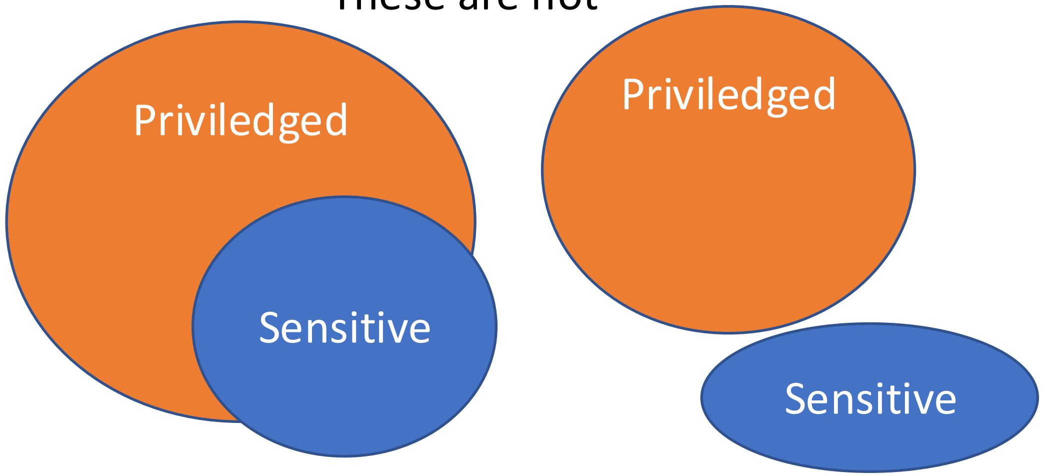
What is required for VMMs to run code directly (in traditional trap-and-emulate)?

- Sensitive instructions: can be run both in user mode and kernel mode
  - Their behavior might change depending on the case
- Priviledged instructions: will result in an interrupt if not in kernel mode

This machine is virtualizable



These are not



This has been taken into account since 2005

- Both from Intel and AMD
- VT technology:
  - The guest OS runs directly on the hardware until it generates an interrupt
  - Which is delivered to the hypervisor
- The hypervisor sets a hardware bitmap to specify all interrupts it wants to be alerted about (trap-and-emulate approach)
- The technique used before that was binary translation:
  - Simply put: replace all unsafe I/O with a trap

Hardware and OSes collaborating, again…

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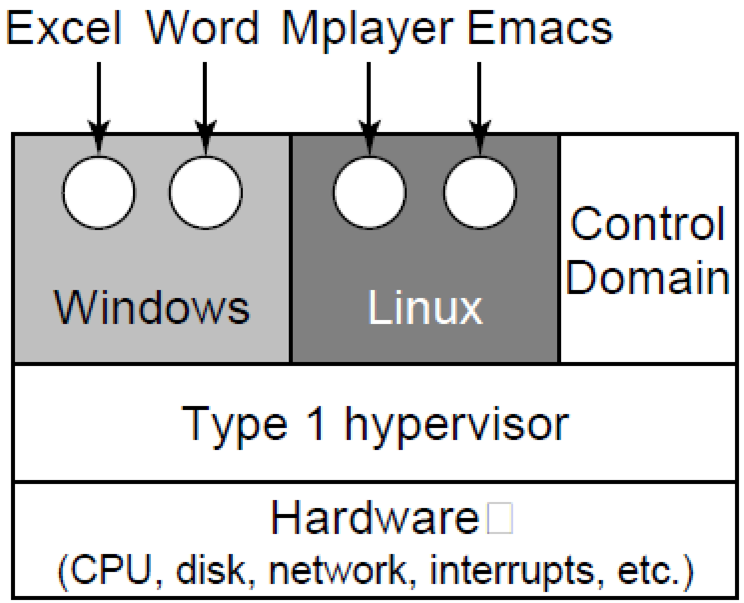
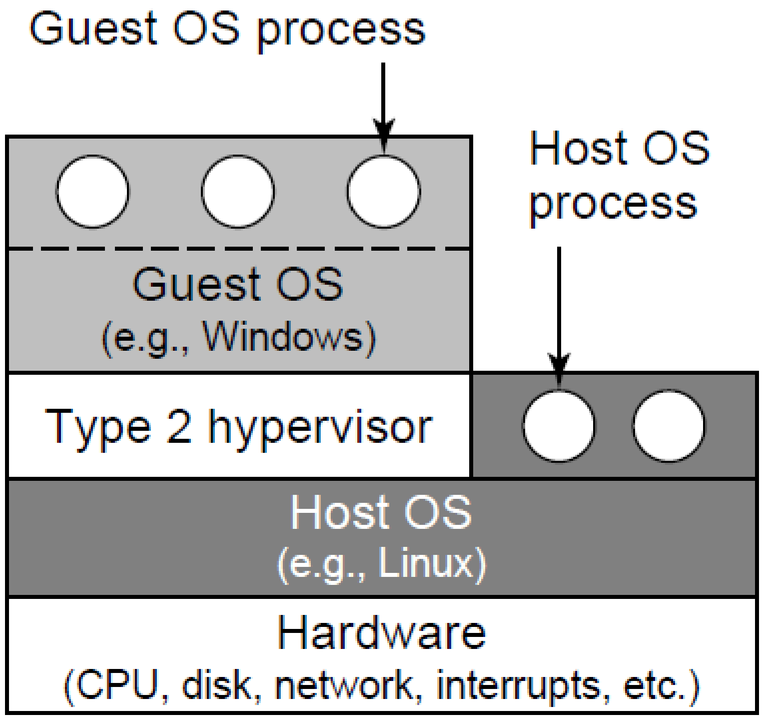
Type 1 and Type 2 Hypervisors



(a)



(b)

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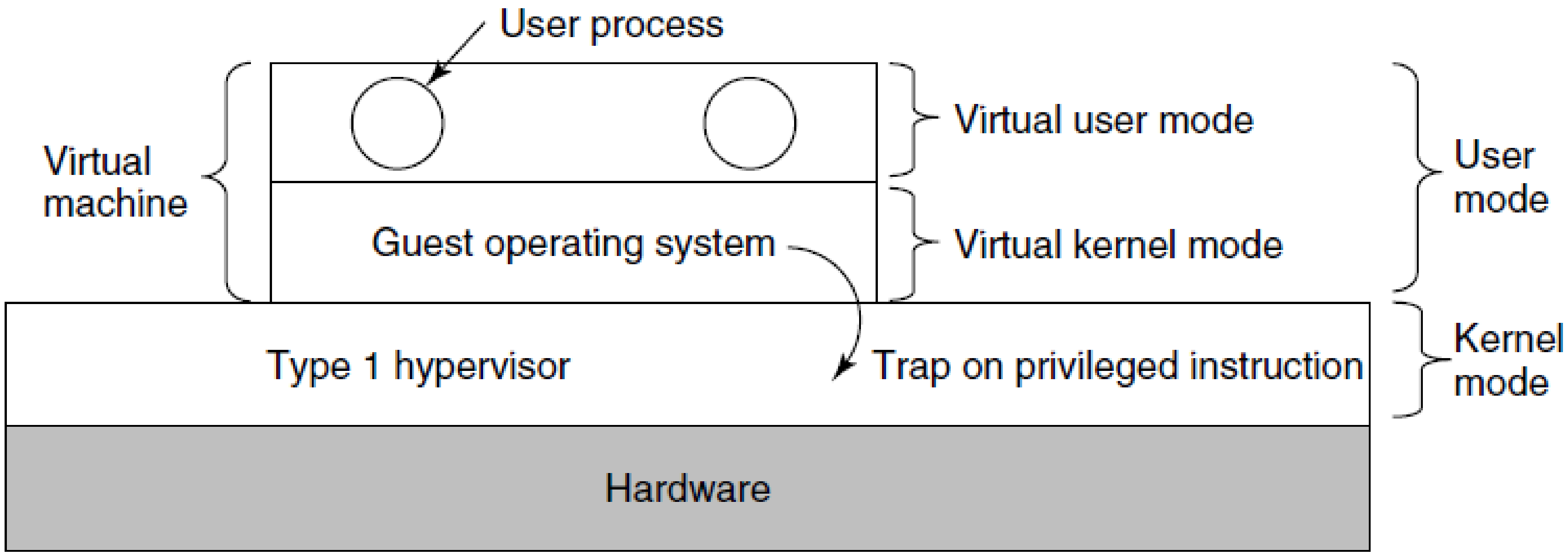
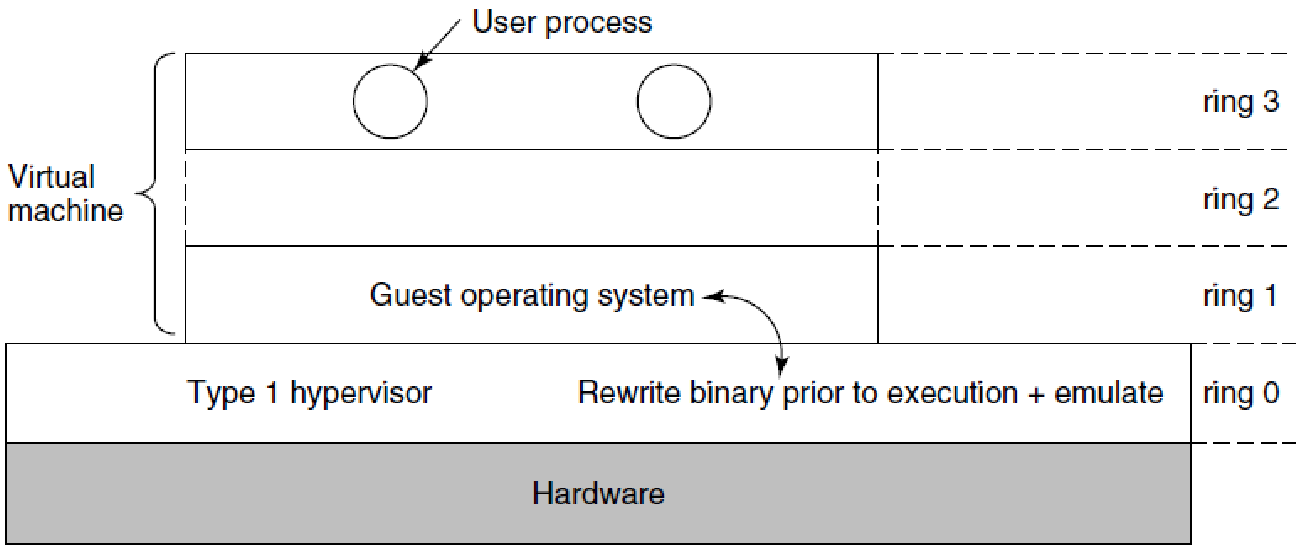
Hypervisor type 1, with VT

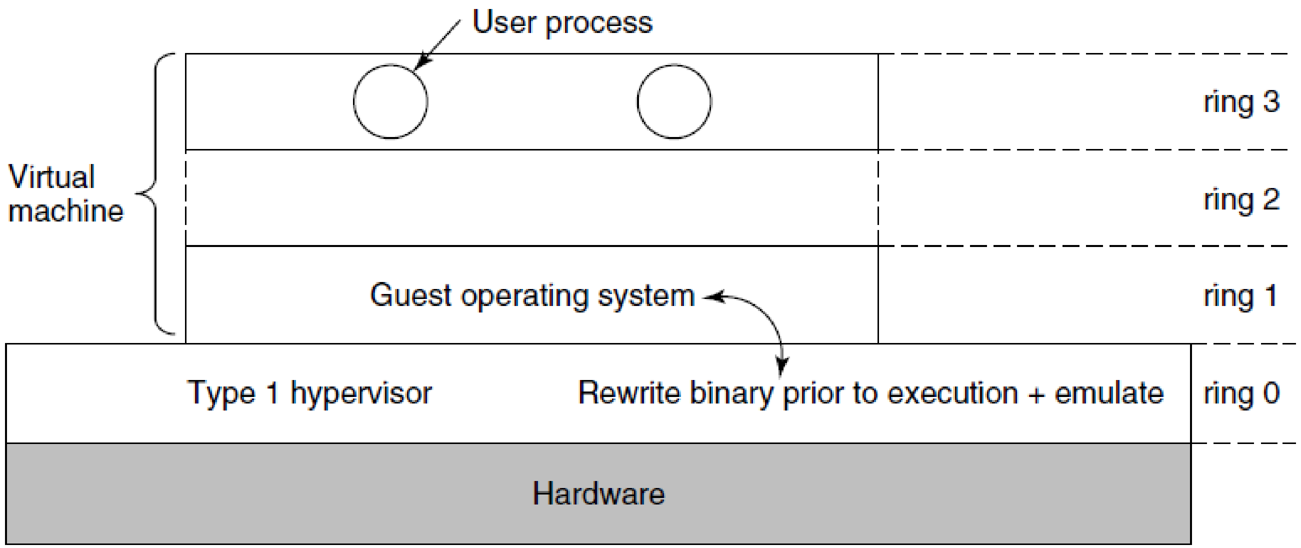


Hypervisor type 1, before/without VT



- Before VT, x86 architectures supported 4 protection modes/rings.
- Without virtualization, OS runs in ring 0, user processes in ring 3
- With virtualization, as shown in figure
- OS privileged instructions will trap, but what about sensitive ones?

Hypervisor type 1, before/without VT



- Sensitive instructions are removed by rewriting the code
- Before executing each **basic block** (a straight sequence of instructions that ends with a branch), the hypervisor scans the block and replaces sensitive instructions

The cost of virtualization

- Traps are expensive
- Because of that, sometimes VT + trap-and-emulate performs worst than binary translation
- In fact, some hypervisor might use binary translation even for privileged instructions
- A small example: Suppose you want to disable interrupts for a while
  - VT: set registers in the hardware
  - Binary translation: rewrite code with an extra if statement

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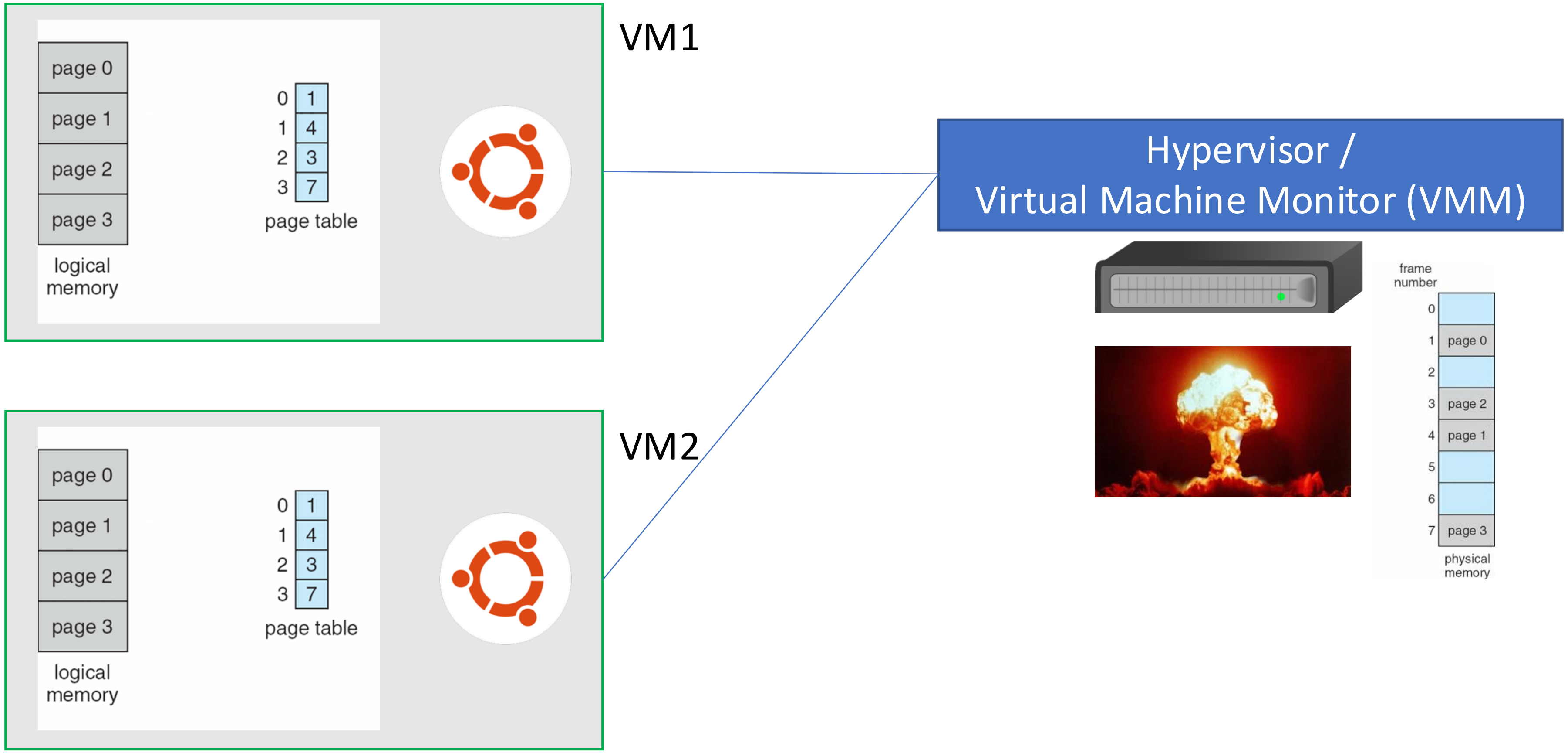
Why do we need to virtualize memory too?



Shadow page table

- The hypervisor maintains a **shadow page table**, to translate each VM’s believed “frame” into the real physical frame
- It is complicated, though:
  - Guest OS page tables and the Hypervisor shadow page table must always be in synch,
  - But the OS can change its page table by simply writing to memory…
  - So, how does the Hypervisor detect a change?
    - Option 1: mark as read-only the VM pages that contain the page table
    - Option 2: do nothing and wait for page faults to happen
      - but this requires extra care of mappings that are removed by the guest OS, because those do not generate page-faults

Page faults

- Guest-induced page faults
  - This are detected by the hypervisor
  - But need to be reinjected to the guest OS
    - Example: OS trying to access a page that has been swapped out of RAM
- Hypervisor-induced page faults
  - Used to keep the shadow page table in synch with the guest OS page table

- When a page faults results in the hypervisor gaining control, we use the term **VM exit**

Reclaiming memory

- If the hypervisor “lies” to the guest OSes pretending it has way more frames it has,
- and then needs to reclaim some of them from OS A and give them to OS B,
- how can it do that?

- If it simply takes them away, it needs to update the OS page tables, but, where are they? When can that be done safely?...

Ballooning

- A small module is loaded in each VM (a balloon) as a pseudo device driver connected to the hypervisor
- The hypervisor can inflate is requested space to force the guest OS in releasing pages from other processes and give them to this module
  - So that the hypervisor can then use them…

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I/O virtualization

- The hypervisor probes devices at startup and
- reports them to the guest OSes.
- When the latter try to use them,
- they will interrupt and the hypervisor can intervene
- BUT: each OS cannot really hold the entire device (e.g., a disk)
- Possible solutions
  - Emulate a disk by means of a file managed by the Hypervisor
  - Emulate network by having the Hypervisor actually behaving as a switch

Single Root I/O Virtualization (SR-IOV)

- For performance, a hypervisor-in-the-middle virtualization approach might not always be optimal
- Modern hardware supports virtualization, which allows for devices to be directly controlled by a guest VM (bypassing the hypervisor)
- Devices with SR-IOV usually have
  - Independent memory space, interrupts and DMA streams for each VM
  - Each VM can map in is own memory its corresponding memory areas

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Other benefits of virtualization - 1

- When a multi-CPU / multi-core machine is used,
- each VM can be assigned a specific subset of cores
- This has an important implication:
    - You could be able to write programs that have a parallelism degree (i.e., number of threads or co-operating processes) that fits the actual number of physical cores the VM you are going to use has access to

Other benefits of virtualization - 2

- Different VMs share memory, and they could potentially be duplicating a lot of (read-only) information
  - Think for instance at having many VMs running the same OS in parallel…
- Deduplication techniques (also referred to as transparent page sharing or content-based page sharing) which are usually used for disks, can be used for memory too!
  - It should not come as a surprise, the physical / logical separation behind VM that is beneficial for running several processes at the same time efficiently can be also beneficial to run several OSes at the same time
  - At the end of the day, the OS runs a collection of processes too

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What exactly is the cloud?

- Public vs private…
- Access to physical hardware, access to virtual OSes, access to specific software
- Big machines / small machines
- Desired characteristics
  - On-demand self-service
  - Broad network access (available via network with standard mechanisms)
  - Resource pooling
  - Elasticity
  - Measured service (also in connection to $)

Types of services

- IAAS (Infrastructure as a Service)
  - Direct access to a specific VM, the user does what (s)he wants
- PASS (Platform as a Service)
  - Specific environment, with given OS, DataBase, WebServer, …
- SAAS (Software as a Service)
  - Offers specific software

Virtual Machine Migration

- What is the server running a bunch of VMs needs maintenance?
- Intuition: move the VM to a different server…
- Simple approach / but not liked by customers who pay
  - Turn off the VM, move the data to a different server, turn on the VM
  - Long downtime… also, turn off when? Ask the user?
- Live migration
  - Keep copy pages in the back, making the transition from one server to another not noticeable (seamless live migration)
  - As you probably imagine, this is everything but simple…

Checkpointing/Snapshotting

- An advantage of VMs is that they can be paused and copied
- If something bad happens, you can then revert the VM to the previous save
- Incremental snapshotting with mechanisms like copy on write can then allow for lightweight checkpointing
  - Which can be also used in combination with migration

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Containers (motivation/intuition)

- VMs virtualize hardware
- Containers provide OS-level virtualization

  →Each process / set of processes believes it's running in its own OS
- Note, under the hood:
  - Processes share the same OS and hardware. Processes are isolated, but the performance of a set of processes is always somehow affected by what other processes do
  - All processes / sets of processes need to share OS

Containers – What are the ingredients for a container / a virtualized OS?

- File system
- Process identifiers
- User identifiers
- Network interfaces and IP addresses
- Inter-Process Communication (IPC) endpoints (e.g., pipes)
- Memory limits
- CPU limits
- I/O limits

…

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- File system
- Process identifiers
- User identifiers
- Network interfaces and IP addresses
- Inter-Process Communication (IPC) endpoints (e.g., pipes)

Can be managed through namespaces

- Memory limits
- CPU limits
- I/O limits

…

Of a different nature, they require some active monitoring

Example of namespace: Mount namespace

- Gives a process its own view of the filesystem mount hierarchy
- Processes in different mount namespaces can see different mounted filesystems
- Created or managed using mechanisms such as clone(), unshare(), and setns()
- Container runtimes combine mount namespaces with mounts and mechanisms such as pivot_root() or chroot() to provide an isolated root filesystem

What about resources that need to be monitored? → cgroups

- One controller per resource type:
  - Resource 1 → cgroups (in a tree) specify “restrictions”
  - Resource 2 → cgroups (in a tree) specify “restrictions”
  - …
- Each process belongs to exactly one cgroup for each resource of interest, and each resource is managed by a controller
- Example:
  - CPU: c1:50%
  - Memory: m1:2GB
  - Processes A and B should use 50% of CPU: A → c1 in CPU / B → c1 in CPU
  - Process B should not use more than 2GB of memory: B → m1 in Memory

Why are cgroups organized in a tree data structure for a given controller? Why multiple controllers?

- Processes A and B should use 50% of CPU
  - **<u>But what if we want specific quotas for them?</u>**
- Also, assume we have a CPU with 4 cores
  - Pinning A and B to 2 cores→ Processes A and B cannot use more than 50% of CPU

Note: there can be different ways of enforcing a restriction, and they might also conflict…

- Cgroup for cpuset controller: use only core 1
- Affinity of a thread in that cgroup: use only core 2

→The OS will enforce the intersection of the two

  →The thread is effectively blocked

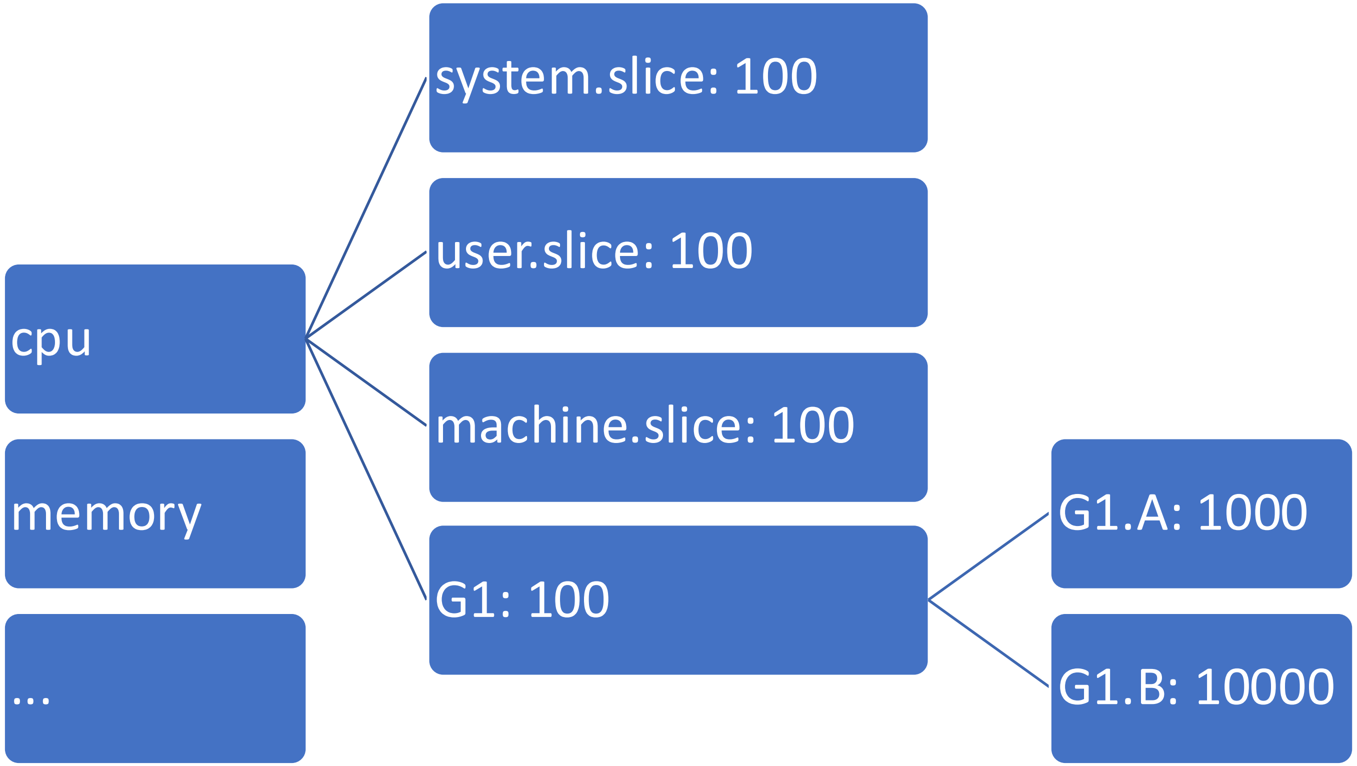
cgroups are organized by controller (subsystem)

- cpu → CPU bandwidth or relative CPU weight
- cpuset → which CPU/NUMA part
- memory → memory usage
- io/blkio → disk I/O
- pids → limits the number of processes/tasks
- devices → access to devices

cgroup v1 and cgroup v2

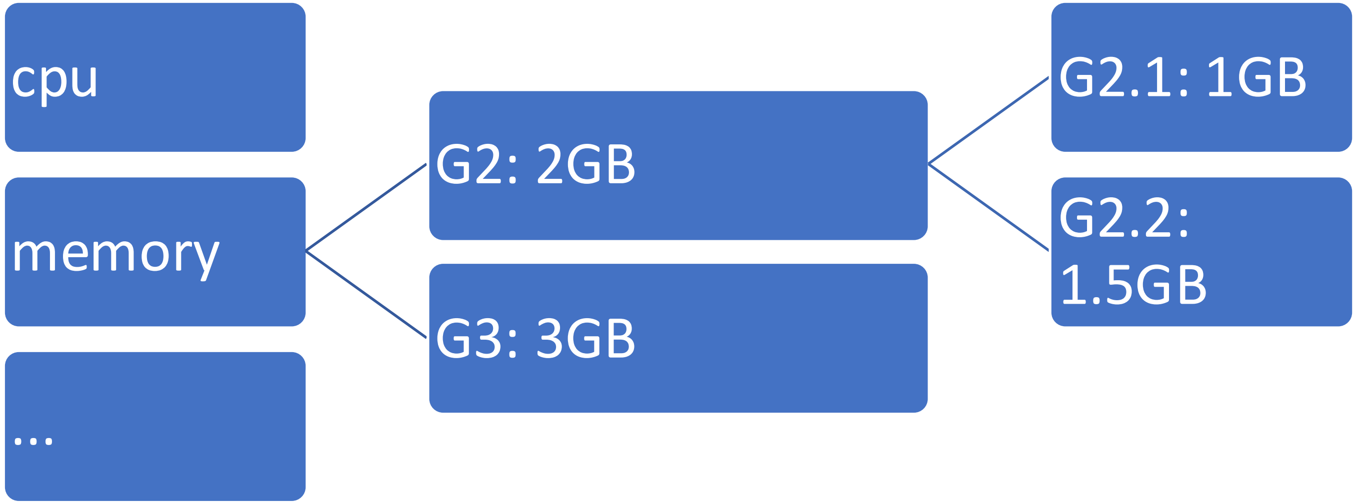
- Please note, the following examples mix v1 and v2 of cgroup
- **cgroup v1**
  - Separate hierarchies may be used for different controllers
  - A process can belong to different cgroups for different controllers
- **cgroup v2**
  - One unified cgroup hierarchy
  - A process belongs to one cgroup in that hierarchy
  - Controllers are enabled within the hierarchy

cgroups are organized by controller (subsystem)



- cpu is defined by means of shares
- Usually:
  - system.slice for system services
  - user.slice for user processes
  - machine.slice for VMs/Containers
- In the example, under full contention, G1 receives approximately 0.25 of the CPU time, G1.A gets 1/11*0.25=0.02 of resources

cgroups are organized by controller (subsystem)



- memory is defined in absolute value
- Processes in G2.1 cannot use more than 1 GB
- Processes in G2.2 cannot use more than 1.5 GB
- Processes in G2.1 together with processes in G2.2 cannot use more than 2GB

Each process belongs to one cgroup in each hierarchy

cat /proc/4042126/cgroup

12:cpuset:/ → cpuset controller, in the root group (/)
11:perf_event:/ → performance monitoring events, root group
10:devices:/user.slice → limited by device access rules of user.slice
9:memory:/user.slice/user-1009.slice/session-9277.scope → memory limits per user/session
8:cpu,cpuacct:/user.slice → CPU shares/usage accounting under user.slice
7:blkio:/user.slice → block I/O controller, user.slice group
6:rdma:/ → RDMA resources, unrestricted (root)
5:pids:/user.slice/user-1009.slice/session-9277.scope → PID count limits per session
4:hugetlb:/ → hugepage allocation control, root group
3:net_cls,net_prio:/ → network classification, root
2:freezer:/ → process freezing, root
1:name=systemd:/user.slice/user-1009.slice/session-9277.scope → systemd bookkeeping
0::/user.slice/user-1009.slice/session-9277.scope → cgroup v2 unified hierarchy

You can find information about each controller

```
ls -la /sys/fs/cgroup/blkio/user.slice/
total 0
drwxr-xr-x 2 root root 0 Oct 10 14:20 .
dr-xr-xr-x 5 root root 0 Jul 18 21:47 ..
--w------- 1 root root 0 Oct 14 06:22 blkio.reset_stats
-r--r--r-- 1 root root 0 Oct 14 06:22 blkio.throttle.io_service_bytes
-r--r--r-- 1 root root 0 Oct 14 06:22 blkio.throttle.io_service_bytes_recursive
-r--r--r-- 1 root root 0 Oct 14 06:22 blkio.throttle.io_serviced
-r--r--r-- 1 root root 0 Oct 14 06:22 blkio.throttle.io_serviced_recursive
-rw-r--r-- 1 root root 0 Oct 14 06:22 blkio.throttle.read_bps_device
-rw-r--r-- 1 root root 0 Oct 14 06:22 blkio.throttle.read_iops_device
-rw-r--r-- 1 root root 0 Oct 14 06:22 blkio.throttle.write_bps_device
-rw-r--r-- 1 root root 0 Oct 14 06:22 blkio.throttle.write_iops_device
-rw-r--r-- 1 root root 0 Oct 14 06:22 cgroup.clone_children
-rw-r--r-- 1 root root 0 Oct 10 16:53 cgroup.procs
-rw-r--r-- 1 root root 0 Oct 14 06:22 notify_on_release
-rw-r--r-- 1 root root 0 Oct 14 06:22 tasks
```

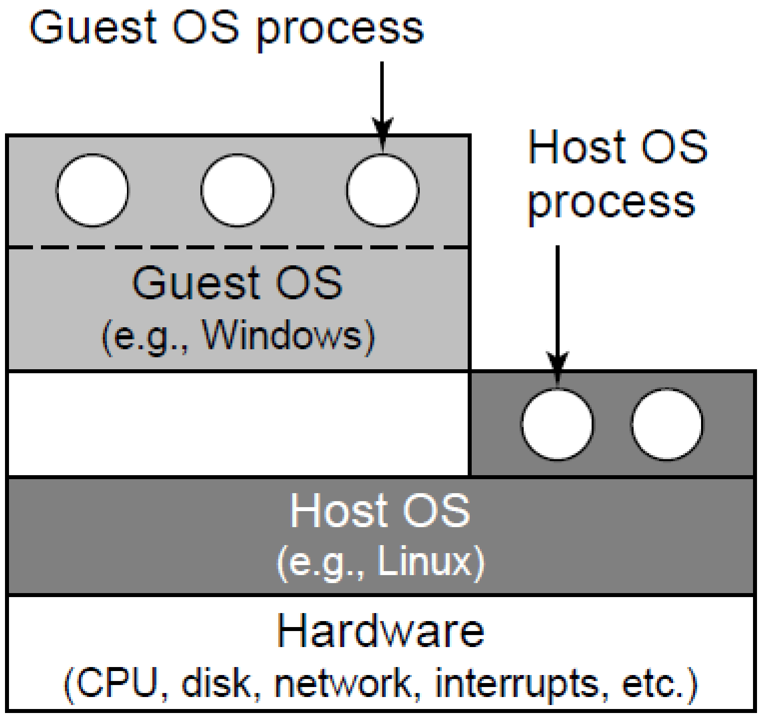
cat /sys/fs/cgroup/blkio/user.slice/blkio.throttle.io_service_bytes would show the accumulated bytes read/written by this cgroup.

Questions

For classic trap-and-emulate virtualization, why must every sensitive instruction also be privileged?

1. Otherwise, the guest OS cannot determine which process issued the instruction.
2. Otherwise, a guest OS could execute a sensitive instruction without trapping, so the hypervisor might not regain control.
3. Otherwise, the hardware cannot distinguish a host OS from a guest OS.

What hypervisor type is shown in the figure?



1. Type 1
2. Type 2
3. Cannot say

Which statement correctly describes a shadow page table?

1. It is the ordinary page table used by the hypervisor when the hypervisor executes its own code.
2. It is maintained by the hypervisor on behalf of a VM and maps the VM's view of memory to real physical frames.
3. It is maintained entirely by the guest OS and is never updated by the hypervisor.

A hypervisor can map the same physical memory page containing identical read-only operating-system code into two different guest VMs.

1. True
2. False

Which mechanisms are used by Containers to control process identifiers?

1. Namespaces
2. Cgroups
3. Both namespaces and cgroups